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Tin metallogenesis associated with granitoids in the southwestern Sanjiang Tethyan Domain: Nature, deposit types, and tectonic setting ☆

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ABSTRACT

Tin mineralization in the southwestern Sanjiang Tethys Sn metallogenic domain extends from the Southeast Asia tin belt in the south. The eastern part of the domain does not contain significant Sn deposits, whereas the central part of the domain, composed of the Changning–Menglian Sn belt and related to ca. 239–178 Ma S-type granitoids generated by the collision between Tengchong–Baoshan and Simao Blocks carries widespread the hydrothermal vein type mineralization. The western part of the domain, composed of the Tengchong–Baoshan Sn metallogenic belt, is related to both 136–113 Ma and 89–52 Ma S-type granitoids. The former granitoids resulted from the collision between Western Burma and Tengchong–Baoshan Blocks, and generated skarn-type tin deposits. The younger groups correspond to the India–Asia continental collision and are associated with the Xiaolonghe and Lailishan greisen type Sn deposits.

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1. Introduction

Tin mineralization in the southwestern Sanjiang Tethys (SWST) Sn metallogenic domain extends from the Southeast Asia tin belt in the south (Fig. 1). The SWST is one of the important branches of the eastern Tethyan tectonic belt (Fig. 2), and thus bears significance for the development of the Tethyan belt within SE Asia (Chen et al., 2004, 2005, 2006; Deng et al., 2010a, 2010b, 2010c). Tectonically, the SWST is considered as a relict continental fragment separated from Gondwana and sandwiched between the Eurasian and the Indian plates. The Sn-polymetallic deposits in the SWST have attracted attention in terms of ore genesis and economic potential (Shi et al., 1989; Hou et al., 2007, 2008). However, exploration and research activities have largely focused on individual deposits and metallogenic regularity, and no detailed regional evaluation has been attempted (e.g., Chen, 1987; Sengor, 1987; Shi et al., 1989; Liu et al., 1993; Hou et al., 2007, 2008). In this paper, we present an overview and re-evaluation of tin metallogenesis in the SWST with regard to: (1) magmatic suites, (2) spatio-temporal distribution, and (3) the nature, deposit types, and tectonic setting of the Sn deposits. We also attempt to compare the SWST Sn metallogenic domain to the Southeast Asia tin belt in order to refine our understanding of Sn metallogeny in the region, which could provide a useful basis for future exploration.

2. Tectonic history of the SW Sanjiang Tethys

The geological evolution of SE Asia during the Paleozoic to Mesozoic defines one of the major Phanerozoic accretionary processes on the globe, related to the closure of the Tethys Ocean, and involving the amalgamation of several micro-continents or continental blocks into the Eurasian continent (Fig. 2; Sengor, 1984; Metcalfe, 1996; Wopfner, 1996; Ueno, 2000; Metcalfe, 2000; Wang et al., 2013; Lai et al., 2013; Goldfarb et al., 2013). The major continental blocks in the SWST separated by tectonic sutures (Figs. 1, 2) are briefly summarized below.

2.1. Continental blocks and arc terranes

2.1.1. Simao Block

The Simao Block is connected to the Indochina Block to the south (Wang et al., 2000; Jin, 2002; Metcalfe, 2002; Lan et al., 2003; Chen et al., 2004; Deng et al., 2012), and traced to the "North Qiangtang Block" or "Eastern Qiangtang Block" to the north (Fig. 1; e.g., Jin, 2002; Zhang et al., 2002; Bian et al., 2004). However, Sone and Metcalfe (2008) regarded this terrane as a probable extension of the Indochina Block. The Simao Block is bound to the west by the Jinghong Suture Zone and to the east and south by the Jinshajiang and Ailaoshan Suture Zones (Fig. 1). The basement rocks of this block are buried beneath thick Paleozoic–Mesozoic sequences. The oldest sedimentary rocks that crop out in this block are Lower Ordovician low-grade metasedimentary

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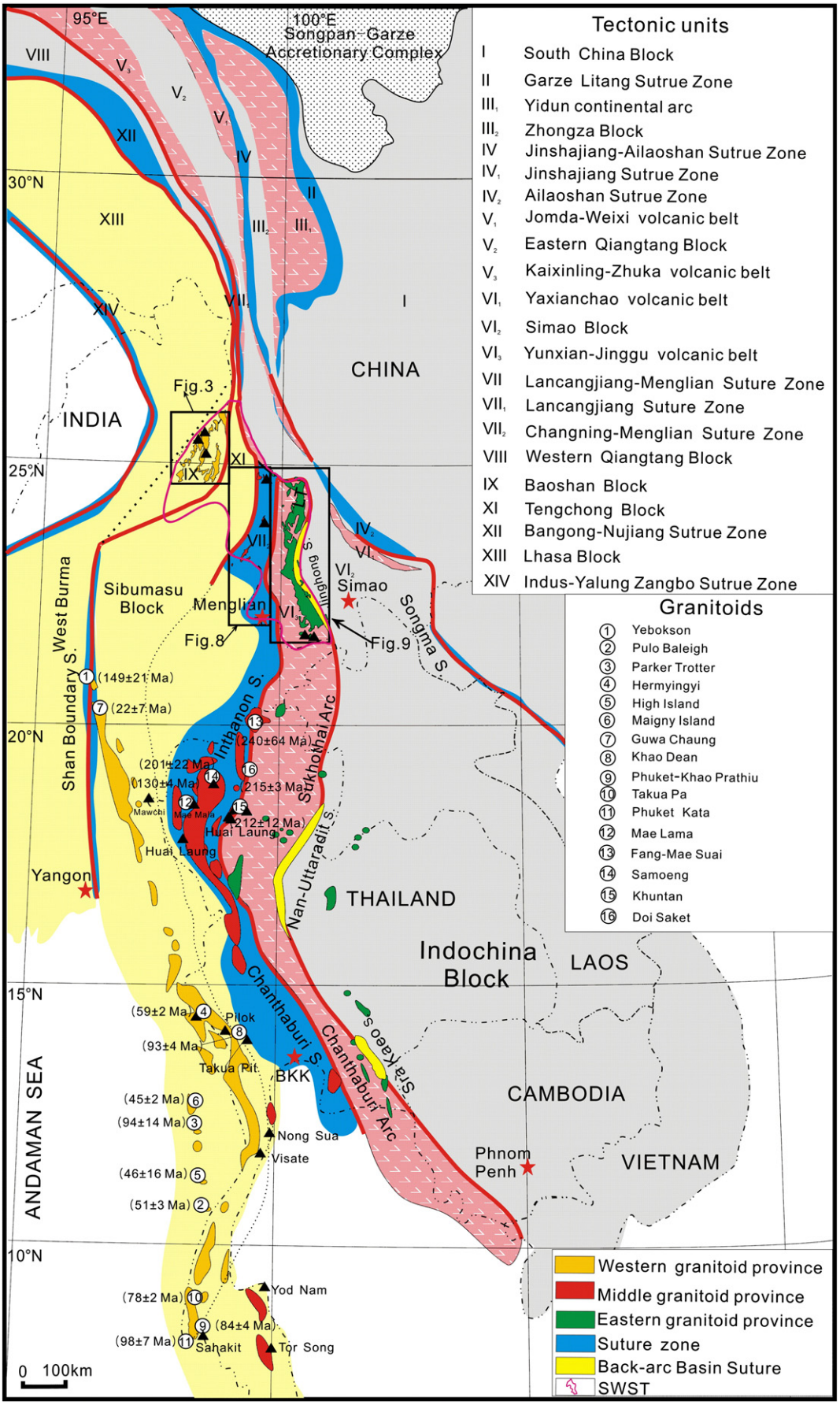


Fig. 1. Simplified geological map showing tectonic framework and granitoids in the SWST and its adjacent areas.
Modified after Shi et al., 1989; Linnen and Williams-Jones, 1995; Yokart et al., 2003; Hou et al., 2007; Sone and Metcalfe, 2008.

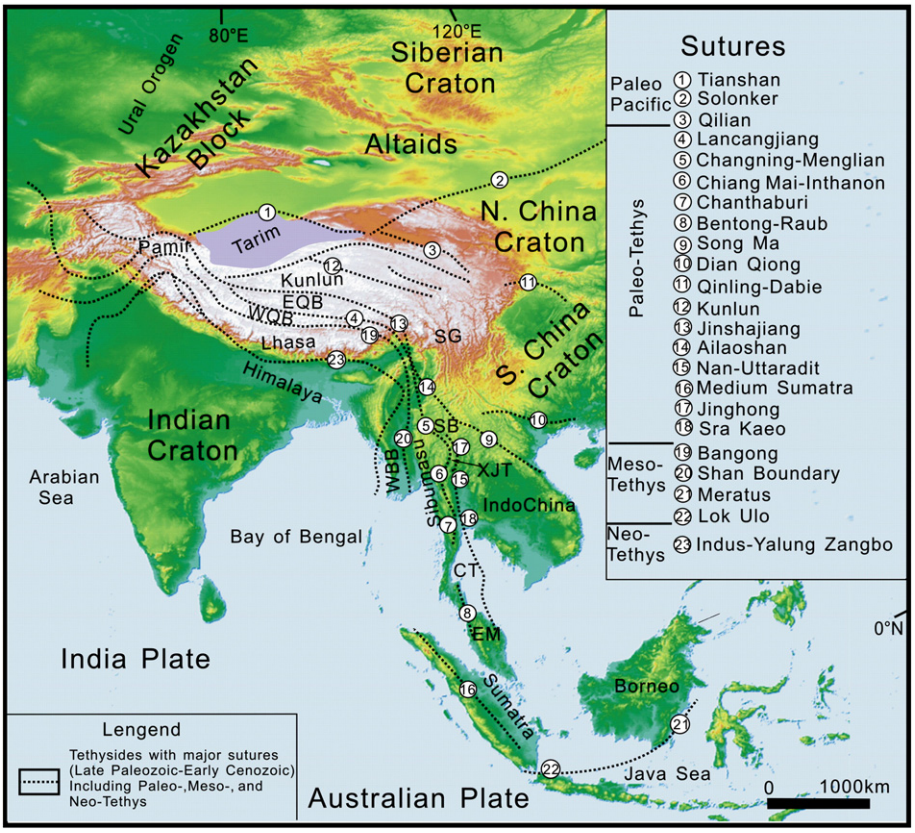


Fig. 2. Tethysides with major sutures from the Late Paleozoic–Early Cenozoic including Paleo-, Meso-, and Neo-Tethys in the SWST and its adjacent areas. WBB = West Burma Block; WQB = West Qiangtang Block; EQB = East Qiangtang Block; SB = Simao Block; SG = Songpan Ganzi accretionary complex; XJT = Yunxian-Jinggu arc Terrane; CT = Chanthaburi arc Terrane; EM = East Malaya arc Terrane.
Modified after www.GoogleMap.Com; Sengor, 1987; Metcalfe, 1996, 2002.

rocks including slates, phyllites, quartzites and meta-limestones. High-grade metamorphic rocks represent continental fragments, and the amphibolites from these units show Sm–Nd whole-rock isochron age of 1437 ± 17 Ma (Zhong, 1998; Wang et al., 2000). Recently, whole rock Rb–Sr age of 236 ± 32 Ma (Mou and Yu, 2002) has been reported for the rhyolite in the Pantiange Formation. The Ordovician unit is unconformably overlain by Middle Devonian to Triassic basal conglomerates, shallow marine sediments, paralic and continental sediments. Mesozoic red beds overlie these sediments.

2.1.2. Yunxian–Jinggu Arc

The Yunxian–Jinggu (YJ) Arc in Yunnan is defined as the northern extension of the Sukhothai Arc. The arc is equivalent to the Lincang arc (e.g. Li, 1988; Liu et al., 1991; Yang, 1998; Ueno, 1999; Barr et al., 2000; Sone and Metcalfe, 2008). It has a continental basement and is bound to the west by the Changning–Menglian (CM), and its eastern boundary is marked by the Jinghong back-arc basin (Fig. 1). The arc was constructed in the Late Carboniferous on the margin of the Simao Block through eastward subduction of the CM Tethyan Ocean. It was separated by the Jinghong back-arc spreading in the Late Carboniferous–Middle Permian and was then accreted back onto the Simao Block by back-arc collapse in the Triassic. Volcanic and volcaniclastic rocks of possible Permo-Triassic ages are distributed extensively in the Yunxian–Jinggu Arc Terrane (e.g., Zhang et al., 1993; Wang et al., 2012; Zhu et al., 2013). Some mafic–ultramafic intrusions have also been reported, such as the Banpo intrusions with zircon U–Pb ages of 298–285 Ma (Jian et al., 2009; Li et al., 2012) and the Nanlinshan intrusions with zircon U–Pb ages of 292 Ma (Hennig et al., 2009). The granitoids are represented by the Lincang granodiorite with zircon U–Pb zircon age of 262 Ma (Yu et al., 2003) and the Jinghong granodiorite with zircon U–Pb zircon ages of 284–283 Ma (Hennig et al., 2009).

As shown in Fig. 1, the Yunxian–Jinggu Arc Terrane was intruded by the Late Triassic to Early Jurassic Lincang biotite granite/monzogranite pluton, with ages of 239–189 Ma (Shi et al., 1989; BGMRY, 1990; Qin, 1991).

2.1.3. Tengchong–Baoshan Block

The eastern boundary of the Tengchong–Baoshan (TB) Block (Fig. 3; Chen et al., 2005) has traditionally been regarded as the Changning–Menglian (CM) Suture in western Yunnan (Metcalfe, 1988; Fang et al., 1994; Fang and Feng, 1996; Hada et al., 1999; Singharawarpan and Berry, 2000; Zhong, 2000). There is a general consensus that the Baoshan Block represents the northern part of the Sibumasu Block. However, the tectonic environment to which the Tengchong Block belongs is controversial, with some workers correlating it to the Lhasa Block (Fan, 1978; Powell and Johnson, 1980) whereas others proposing a connection with the Sibumasu Block (Sengor, 1979, 1984; Metcalfe, 1984, 1988; Cong et al., 1993; Metcalfe, 2000, 2002; Y.K. Shi et al., 2008). Although the Tengchong Block has some distinct features, it shares a number of similarities with the Sibumasu Block as follows. (1) At the end of the Sakmarian stage of the Early Permian, the elongate Cimmerian continental strip (Sengor, 1984) separated from eastern Gondwana, including the Baoshan Block and possibly the Tengchong Block (Wopfner, 1996), and the Sibumasu Block (Metcalfe, 1984). However, the Lhasa Block was part of the separating eastern Cimmerian continent, but it is recognized that the Lhasa Block docked later onto Eurasia in the Late Jurassic/Early Cretaceous (Metcalfe, 1988). (2) The Gaoligong Formation in the Tengchong Block is regarded as the oldest basement rocks (1094–840 Ma; Zhong, 1998), and comprises metamorphosed volcano-sedimentary rocks, tonalite and gabbro. These Proterozoic rocks are similar to the Mogok rocks in the Sibumasu Block in northern

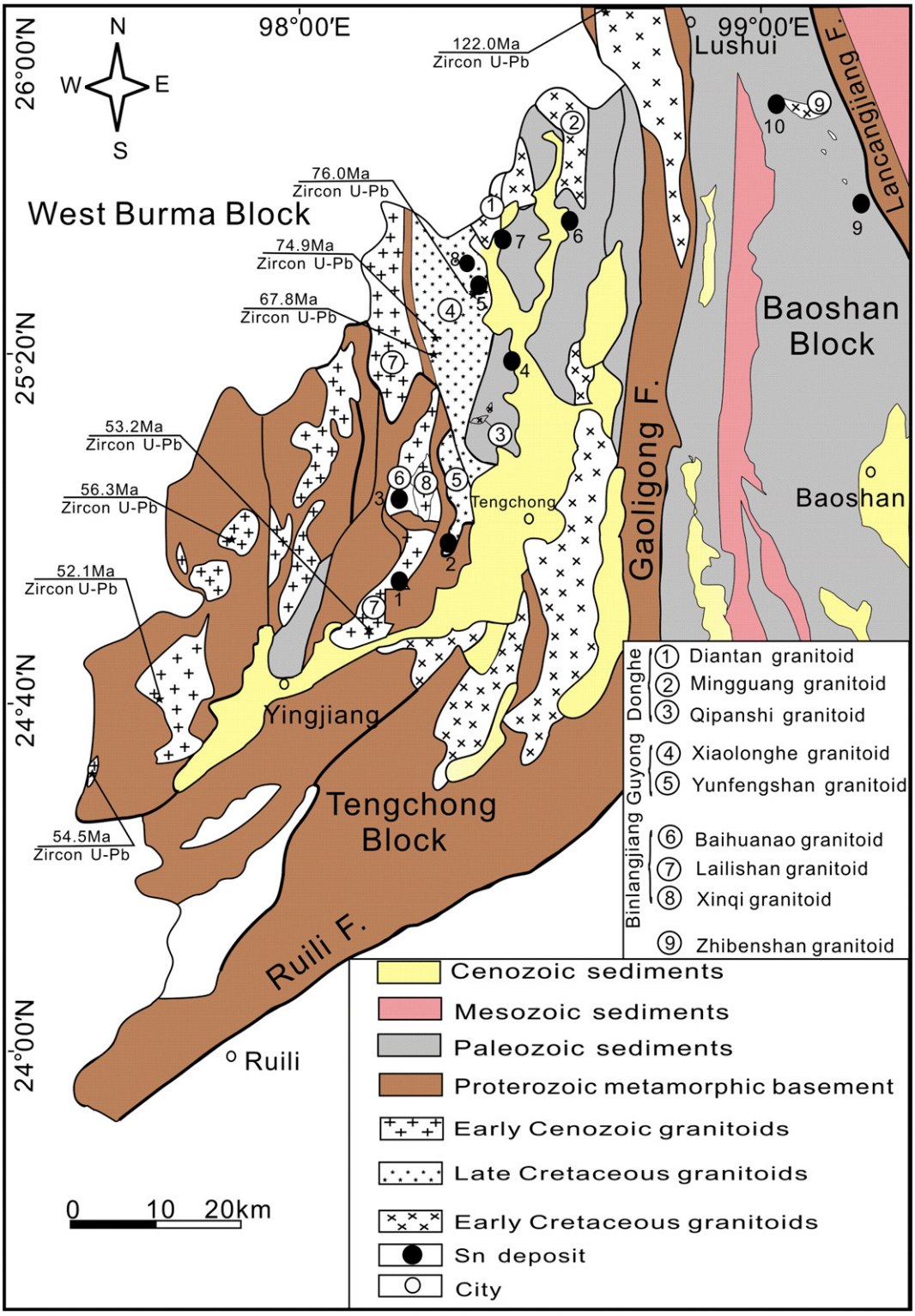


Fig. 3. Simplified geological map showing the granitoids and the distribution of significant Sn deposits in the Tengchong–Baoshan Sn metallogenic belt. 1 = Lailishan; 2 = Laopingshan; 3 = Baihuanao; 4 = Tieyaoshan; 5 = Xiaolonghe; 6 = Dadongchang; 7 = Daintan; 8 = Dasongpo; 9 = Tiechang; 10 = Shiganghe.
Modified after Hou et al., 2008; Yang et al., 2009; Xu et al., 2012.

Burma (Chen, 1991; Cong et al., 1993; Fan and Zhang, 1994; Zhong, 1998). Metcalfe (2002) suggested that the differences in stratigraphy between the Tengchong and Baoshan Blocks, with some volcanics and red beds present in Baoshan, but not in Tengchong, do not necessarily indicate a significantly different tectono-stratigraphic setting. Both Permian sequences in these blocks represent the sediments deposited in different parts of a Permian rift system environment. (3) Late Paleozoic Gondwana faunas and floras in the Tengchong Block share similar features with contemporary faunas and floras in the Baoshan Block and Sibumasu Terrane up until the Sakmarine (Wopfner, 1996; Wang and Wang, 2001; Y.K. Shi et al., 2008). (4) These eastern Cimmerian Blocks are characterized by Late Carboniferous–Early Permian glacial-marine

diamictites, which are interbedded with marine clastics and turbidites (Wopfner, 1996; Wang and Wang, 2001). (5) More recently, based on tectonic and geophysical evidence, the TB Block has been considered as a segment of the Gondwana supercontinent (e.g., Fan and Zhang, 1993; Metcalfe, 1996; Wopfner, 1996; Zhong, 1998; Pan et al., 2004).

2.2. Suture zones

2.2.1. Paleo-Tethys Suture Zones

The Paleo-Tethys is principally represented in SE Asia by the Lancangjiang, CM, Nan–Uttaradit, Sra Kaeo, Bentong–Raub, Jinshajiang, Ailaoshan, and Song Ma Suture Zones (Fig. 1). These suture zones include accretionary complexes containing fault-bound packages of ocean floor sequences, as pillow basalts, ribbon-bedded cherts, pelagic limestones, shallow-marine limestones, siliceous mudstones and turbidite 'flysch' sediments. Ages of oceanic deep-marine ribbon-bedded cherts of the Paleo-Tethys range from late Early Devonian to Middle Triassic (Metcalfe, 1997, 2002).

In the western Yunnan province of China, the CM Suture Zone is generally regarded as the main Paleo-Tethys Suture Zone (Liu et al., 1991; Ueno et al., 2003). To the south it is correlated with the Inthanon Suture Zone in Thailand, and to the north it is correlated with the Lancangjiang Suture in the eastern Tibet (Fig. 1). The oldest deep-sea sediment bears Devonian *Eoalbaillella lilaensis* radiolarians in the Lalei Formation in the Menglian area (Feng and Liu, 1993; Feng and Ye, 1996; Feng et al., 1997). The chronostratigraphic age range of pelagic sediments in the CM Suture Zone is from late Middle Devonian cherts with the *Entactinia–Entactinisphaera* radiolarian assemblage to the Middle Triassic *Triassocampe deweveri* radiolarian assemblage (e.g., Feng and Ye, 1996; Feng et al., 2001). Ages in the range of 349–331 Ma were reported based on single zircon U–Pb dilution method (Duan et al., 2006) for the MORB-type ophiolite assemblages. Thus, the opening of the CM Tethyan Ocean is considered to have occurred during the Middle Devonian, as also constrained from radiolarian data. The closure of the CM Tethyan Ocean occurred during the Middle Triassic, as inferred from the sedimentary records of the Shanglan Formation in the Simao Block. The CM Tethyan ocean lithosphere was subducted eastward under the Simao Block in the Late Carboniferous (e.g., He and Liu, 1993; Ueno et al., 2003; Yang et al., 2007; Feng et al., 2008; Sone and Metcalfe, 2008). Recently, SHRIMP U–Pb zircon age of 267.1 ± 3.1 Ma (Jian et al., 2009) has been reported for the Menglian ophiolites from the Damangguangfang. Wei et al. (2008) reported SHRIMP U–Pb zircon ages of 297 ±5 Ma for the Jicha ophiolite and 303.4 ± 4.4 Ma for the E'za grabbro from Baijixun mafic–ultramafic rocks. Based on the ages of ophiolites (Wei et al., 2008; Jian et al., 2009), the CM Suture Zone is regarded as a supra-subduction zone (Zhang et al., 1985; Wu et al., 1995; Zhang, 2000; Jian et al., 2009). The collision between Tengchong–Baoshan and Simao Blocks resulted in the formation of the CM Suture and generated a series of S-type granitoids associated with the hydrothermal vein type Sn deposits.

The Jinghong Back-arc Basin Suture is a minor suture between the Simao Block and the Yunxian–Jinggu arc terrane (Fig. 1). The suture is equivalent to what has been previously referred to as the Lancangjiang Belt or the southern Lancangjiang Suture by some authors (e.g., Liu et al., 1991; Fang et al., 1994; Chen et al., 1995). However, the term Lancangjiang Suture should be restricted to Tibet. Accordingly, the (southern) Lancangjiang Suture in Yunnan, which is not attributable to the Paleo-Tethys Suture, is re-named the Jinghong Suture by Sone and Metcalfe (2008). The Jinghong Back-arc Basin opened in the Permian when the Yunxian–Jinggu arc separated from the margin of Simao Block, and closed in the Late Triassic (Sone and Metcalfe, 2008). The deep-sea cherts were dated by radiolarians as late Early Permian from the upper Longdonghe Formation (Feng et al., 2000), and late Middle or earliest Late Permian red cherts of the Daxinshan Formation (BGMRY, 1990; Feng and Liu, 1993; Feng et al., 2002). The Daxinshan cherts are associated with tholeiitic basalts in the Jinghong Mélange (Feng et al., 2002). These deep-sea sediments are limited to the Permian, suggesting the existence of a minor short-lived ocean basin.

2.2.2. Meso-Tethys Suture Zones

During the Late Triassic–Late Jurassic, the northward shift of the Lhasa and West Burma Blocks resulted in subduction of the Meso-Tethys and the formation of the Neo-Tethys (Metcalfe, 1996, 2002; Bortolotti and Principi, 2005). The closure of the Meso-Tethys formed a series of suture zones (Figs. 1, 2): the Bangong–Nujiang in Tibet (R.D. Shi et al., 2008), Shan Boundary in Burma, and Lok Ulo and Meratus Sutures in SW Borneo (Metcalfe, 1997, 2002).

The Bangong–Nujiang Suture Zone, which extends for more than 2800 km across the central Tibetan Plateau and western Yunnan and to the south, is correlated with the Shan Boundary in Burma (Figs. 1, 2). The Bangong–Nujiang Tethyan Ocean opened during the Middle Permian, and closed during the Early Cretaceous (Zhu et al., 2013). In Tibet, the Bangong–Nujiang Tethyan Ocean lithosphere was subducted either northward under the Qiangtang Block during the Middle Triassic–Middle Jurassic (Yin and Harrison, 2000; Kapp et al., 2003; Guynn et al., 2006; Kapp et al., 2007) or southward under the Lhasa Block during the Permian (possibly as early as ~260 Ma)–Early Cretaceous (Hsü et al., 1995; Pan et al., 2006; Zhu et al., 2013). In the western Yunnan, the Bangong–Nujiang Tethyan Ocean lithosphere was subducted south-eastward under the TB Block during the Permian–Early Cretaceous. The collision between Western Burma and Tengchong–Baoshan Blocks resulted in the formation of the Bangong–Nujiang Suture and generated S-type granitoids associated with the Daintan skarn type Sn deposit in the TB Block. There is no evidence for the remnants (e.g., ocean floor stratigraphy, mélange, ophiolites) of the Meso-Tethys that must have existed in the western Yunnan. This zone appears to correlate with the Gaoligong metamorphic belt in the western Yunnan that forms the boundary between Baoshan and Tengchong Blocks, and it has been interpreted as a major transcurrent shear zone in the western Yunnan (Socquet and Pubellier, 2005).

2.2.3. Neo-Tethys Suture Zones

The Indus–Yalung Zangbo Ocean opened around Middle Jurassic; its northward subduction resulted in the formation of the Gangdese magmatic arc during the Late Cretaceous, and closed at 50 Ma. The Paleocene India–Asia continental collision formed the Indus–Yalung Zangbo Suture (Hébert et al., 2012; Lu et al., 2012; Figs. 1, 2), and resulted in a series of S-type granitoids associated with the Xiaolonghe and Lailishan greisen type Sn deposits in the TB Block. The Indus–Yalung Zangbo Suture is not present in the western Yunnan, and it is described in detail in Chen et al. (2012), Guan et al. (2012), and Zhu et al. (2013).

3. Granitoid provinces, petrography and magma derivation

The granitoids of the SWST have petrographic and geochronological features which permit them to be grouped into north–south elongate belts (Fig. 1). The western Tengchong–Baoshan (TB) granitoid province is located in the Tengchong–Baoshan Block. The middle Changning–Menglian (CM) granitoid province is located in the CM Suture Zone. The western Yunxian–Jinggu (YJ) granitoid province is located in Yunxian–Jinggu arc terrane from the east by the Jinghong Suture to the west by the CM Suture.

3.1. The Tengchong–Baoshan granitoid province

The Tengchong–Baoshan granitoid province in the SWST is located in the northern segment of the Sibumasu Terrane (Fig. 1). The granitoids are exposed in the TB Block such as the Donghe (Diantan, Mingguang, and Qipanshi), Guyong (Xiaolonghe and Yunfengshan), Binlangjiang (Baihuanao, Xiqi, and Lailishan), and Zhibenshan (Fig. 3). The rocks are composed of monzogranites and granodiorites that predominate over tonalites and quartz diorites (Fig. 4). These granitoids have similar

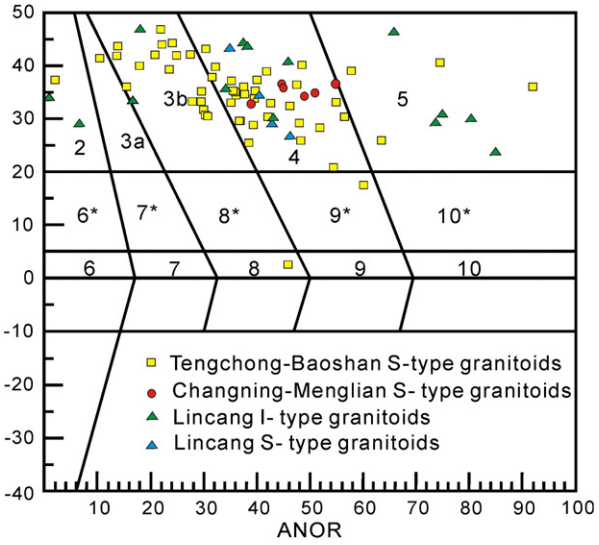


Fig. 4. Q′-ANOR normative composition diagram of granitoids in the SWST (after Streckeisen and Le Maitre, 1979). Q′ = Q × 100 / (Q + Or + Ab + An), ANOR = An × 100 / (Or + An). 2 = alkali feldspar granite; 3a = syenogranite 3b = monzogranite; 4 = granodiorite; 5 = tonalite; 6* = alkali feldspar Qz-syenite; 7* = Qz-syenite; 8* = Qz-monzonite; 9* = Qz-monzodiorite; 10* = Qz-diorite; 6 = alkali feldspar syenite; 7 = syenite; 8 = monzonite; 9 = monzodiorite; 10 = diorite.
All data are from Yu et al. (2003), Hou et al. (2008), Yang et al. (2006, 2009), Li et al. (2011, 2012), Kong et al. (2012), and Nie et al. (2012).

major element contents, with high SiO_2 (average 71.81%) and low MgO, CaO and P_2O_5. They have relatively high Al_2O_3 contents (average 14.31%) and K_2O contents (average 4.70%), which define the granitoids as members of the high-K (calc-alkaline) series (Fig. 5) (Yang et al., 2006; Hou et al., 2008; Yang et al., 2009). Their alumina-saturation index (A/CNK molar) varies about 1.17 and their A/NK molar is about 1.52. These characteristics and corundum in the CIPW norm (>1 wt.%) define the TB granitoids as peraluminous (Fig. 6a). The Al vs. (Na + K + 0.5Ca) diagram (Fig. 6b) shows that TB granitoids have characteristics of S-type granites. The granitoids have similar chondrite-normalized REE patterns, showing high REE concentrations (average 242.48 ppm), chondrite-normalized La/Yb ratios varying about 15.33 and chondrite-normalized La/Sm ratios varying about 6.60. Their chondrite-normalized REE patterns show LREE enriched feature with negative Eu anomalies (Eu/Eu* = 0.45) (Yang et al., 2006; Hou et al., 2008; Yang et al., 2009).

The Sr and Nd isotopic compositions of the TB granitoids show a narrow range of εNd(t) values (−14.7 to −8.3) and εHf(t) values (−10.4 to −2.8), whereas their initial $^{87}Sr/^{86}Sr$ (Isr) ratios vary from 0.71040 to 0.71855 (Hou et al., 2008; Yang et al., 2009; Xu et al., 2012). They show Nd model ages of 1.03–1.43 Ga, which are similar to the crustal ages of the Proterozoic basement rocks along the northeastern margin of the TB Block, indicating that they were derived from an older crustal source (Hou et al., 2008).

In the Rb vs. Yb + Ta plots, the TB granitoids mostly fall in syn-collision field and partly in volcanic arc field (Fig. 7a), suggesting a collision-related setting. On the other hand, the Nb vs. Y plots do not yield unambiguous information for the tectonic setting of the formation of the TB granitoids, because they straddle the boundaries between within plate, volcanic arc and collisional granites (Fig. 7b). In the R2 vs. R1 plots, the TB granitoids mostly fall in the syn-collision and post collision fields (Fig. 7c), whereas the Al_2O_3 vs. SiO_2 plots suggest their formation in a post collision-related setting (Fig. 7d). As such, the compositions of the TB granitoids are compatible with crustal thickening in an intra-crustal setting but not in a subduction-related setting.

The magmatism in the TB area is characterized by predominant S-type granitoids with emplacement ages ranging from Early Cretaceous, to Late Cretaceous and Early Cenozoic. Moreover, there is a younging trend oblique to the regional geological strike, from NE to SW. In the first stage, the Early Cretaceous granitoids are related to magmatism from 136 Ma to 113 Ma based on K–Ar ages of 136–113 Ma, Rb–Sr ages of 132–127 Ma, and zircon U–Pb ages of 126–121 Ma (Shi et al., 1989; Zhang, 1996a; Tao et al., 2009; Yang et al., 2009; Xu et al., 2012). The Early Cretaceous granitoids likely resulted from crustal melting, possibly associated with the collision between the west Burma Block and the TB Block in the Early Cretaceous. In the second stage, the Late Cretaceous Guyong granitoid is related to magmatism from 89 Ma to 65 Ma, as inferred from K–Ar ages of 84–65 Ma, Rb–Sr ages of 89–69 Ma, and zircon U–Pb ages of 76–67 Ma (Chen, 1987; Liu et al., 1993; Lu et al., 1993; Xia, 2003; Yang et al., 2009; Jiang et al., 2012). The Early Cenozoic granitoids in the third stage are mainly composed of the Xinqi, Lailishan and Baihuanao plutons emplaced during 60 Ma to 52 Ma, as indicated by K–Ar ages of 58–52 Ma, Rb–Sr ages of 60–51 Ma, and zircon U–Pb ages of 56–52 Ma (NRGST, 1985; Shi et al., 1989; Chen, 1991; Xu et al., 2012). The Late Cretaceous–Early Cenozoic granitoids were generated in the lower to middle crust by partial melting. This partial melting was induced by crustal thickening related to the subduction of the Neo-Tethyan plate and the subsequent collision between the Indian and Asian continents (Yang et al., 2006; Hou et al., 2008; Yang et al., 2009).

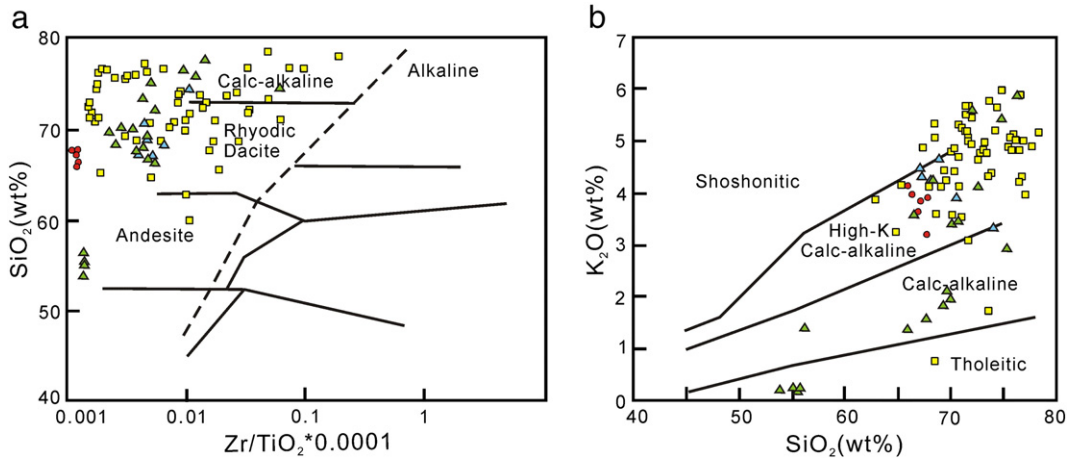


Fig. 5. (a) Zr/TiO_2 vs. SiO_2 diagram, the dashed line separates calc-alkaline and alkaline compositions (after Winchester and Floyd, 1977); (b) SiO_2–K_2O diagram of granitoids in the SWST. All data and symbols as in Fig. 4.

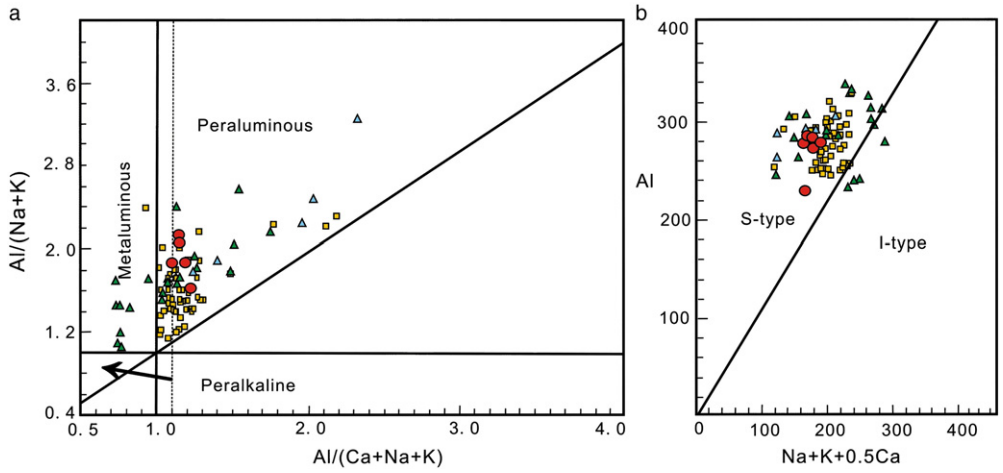


Fig. 6. Classification diagrams of granitoids in the SWST: (a) A/NK–A/CNK (after Maniar and Piccoli, 1989) and (b) Al-(N + K + 0.5Ca) (after Takahashi et al., 1980). A = Al_2O_3, N = Na_2O, K = K_2O, C = CaO (all in molar proportion). All data and symbols as in Fig. 4.

In contrast, biotite granite and hornblende-biotite granite/granodiorite occur in the Sibumasu Block. This block is almost exclusively made up of granitoids with Rb–Sr whole rock isochron ages varying from Cretaceous to Tertiary (Fig. 1), such as 149 ± 21 Ma for Yebokson, 94 ± 14 Ma for Parker Trotter, 59 ± 2 Ma for Hermyingyi, 51 ± 3 Ma for Pulo Baleigh, 46 ± 16 Ma for High Island, 45 ± 2 Ma for Maigny Island, and 22 ± 7 Ma for Guwa Chaung in Burma (Nakapadungrat et al., 1984; Charusiri et al., 1993; Schwartz et al., 1995; Yokart et al., 2003), and 130 ± 4 Ma for Mae Lama, 98 ± 7 Ma for Phuket–Kata, 93 ± 4 Ma for Khao Dean, 84 ± 4 Ma for Phuket–Khao Prathiu,

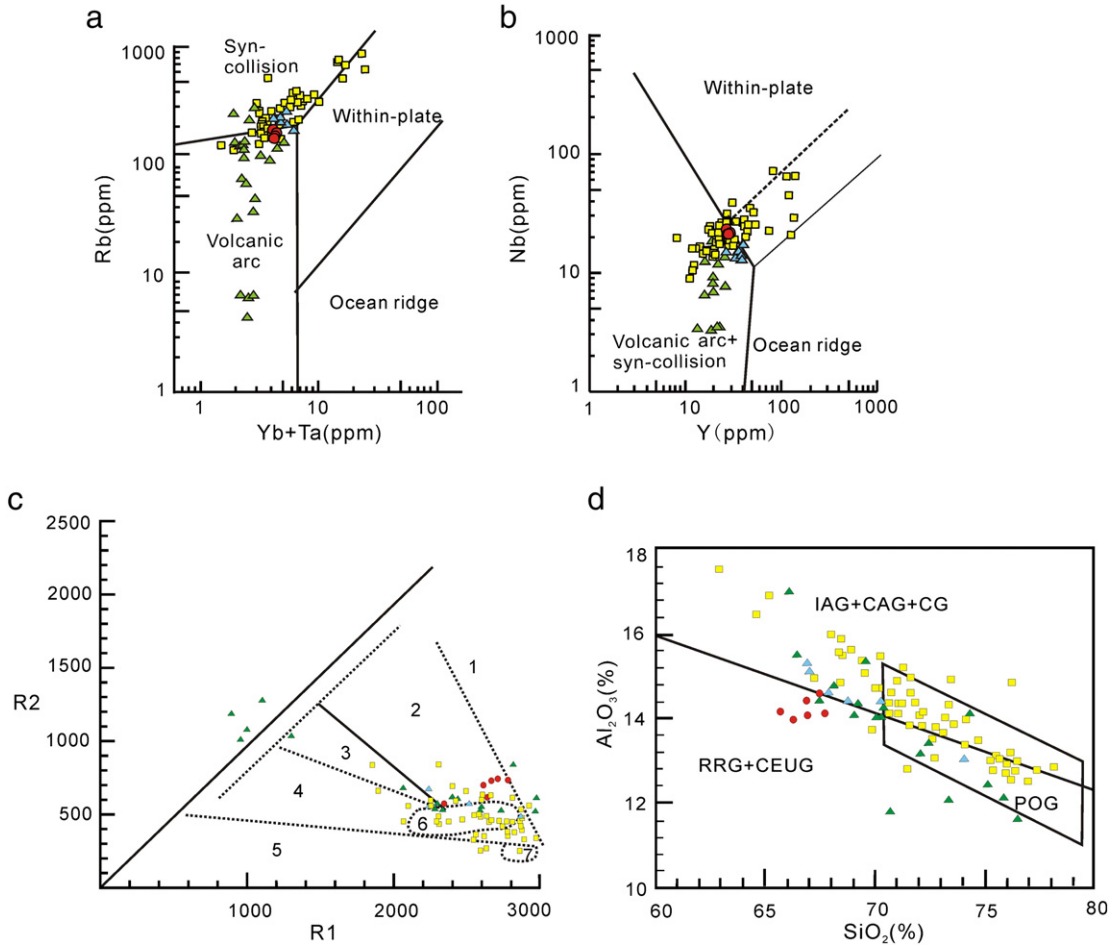


Fig. 7. Discrimination diagrams of granitoids in the SWST: (a) Rb vs. Yb + Ta; (b) Nb vs. Y (after Pearce et al., 1984); (c) R1 vs. R2 (after Batchelor and Bowden, 1985); and (d) SiO_2–Al_2O_3. 1 = Mantle fractionates; 2 = Pre-plate collision; 3 = Post-collision uplift; 4 = Late orogenic; 5 = Anorogenic; 6 = Syn-collision; 7 = Post-orogenic; RRG = rift related granitoids; CEUG = continental epeirogenic uplift granitoids; POG = post orogenic granitoids; IAG = island arc granitoids; CAG = continental arc granitoids; CCG = continental collision granitoids; R1 = 4Si − 11(Na + K) − 2(Fe + Ti); R2 = 6Ca + 2Mg + Al. All data and symbols as in Fig. 4.

and 78 ± 2 Ma for Takua Pa in western Thailand (Darbyshire and Swainbank, 1988; Schwartz et al., 1995). The Western Granitoid Province consists of Late Cretaceous to Middle Tertiary S-type (biotite, two-mica and leucocratic) granites with minor I-type (hornblende–biotite) intrusions in Thailand and Burma (Schwartz et al., 1995).

3.2. The Changning–Menglian granitoid province

The Changning–Menglian granitoid province comprises a series of granitoids including the Lailitoushan, Yunling and Gengma (Fig. 8). These granitoids are dominated by granodiorite (Fig. 4). They have similar major element contents, with high SiO_2 (average 66.82%), and low MgO, CaO and P_2O_5. They have relatively high Al_2O_3 contents (average 14.17%). They belong to high-K (calc-alkaline) series (Fig. 5) and have relatively high K_2O contents (average 3.81%). They are generally peraluminous, with average A/CNK of 1.16 (Fig. 6a). They have high normative corundum of >1%. According to the Al vs. (Na + K + 0.5Ca) diagram, the CM granitoids are S-type granites (Fig. 6b). The granodiorites have similar chondrite-normalized REE patterns, with high REE concentrations (average 251.24 ppm), chondrite-normalized La/Yb ratios ranging from 10.86 to 17.78, and chondrite-normalized La/Sm ratios ranging from 3.98 to 6.05. Their chondrite-normalized REE patterns exhibit LREE enrichment (LREE/HREE = 9.06) with negative Eu anomalies (Eu/Eu* = 0.56), which probably indicate fractional crystallization of plagioclase. The CM granitoids have negative εNd(t) values with average of −14.0, reflecting crustal derivation. They have ancient Nd model ages of 2.7–1.2 Ga, indicating that they were derived from an ancient crustal source (Nie et al., 2012).

The S-type CM granitoids have ages ranging from early Late Triassic to late Early Jurassic (237–178 Ma), such as Haobadi with a U–Pb age of 237 ± 10 Ma, Fangling and Gengma with Rb–Sr whole rock isochron ages of about 210 Ma (Shi et al., 1989; BGMRY, 1990; Nie et al., 2012), and minor Tertiary granitoids with emplacement ages of 50–40 Ma (Shi et al., 1989; Yu et al., 2008; Chen et al., 2010). Some cassiterite–tourmaline deposits (e.g., Ximeng and Haobadi) are related to the S-type granitoids. In contrast, the northern Thailand granite province consists mainly of S-type granitoids such as Fang–Mae Suai, Samoeng, Khuntan, and Doi Saket with Rb–Sr whole rock isochron ages of 240 ± 64 Ma, 201 ± 22 Ma, 212 ± 12 Ma, and 215 ± 3 Ma, respectively (Fig. 1; Nakapadungrat et al., 1984; Cobbing et al., 1986; Charusiri et al., 1993; Schwartz et al., 1995; Yokart et al., 2003).

Geochemical plots of the CM granitoids do not yield unambiguous information about the tectonic setting of their formation (Fig. 7). However, the petrogenesis of the S-type CM granitoids has been explained as due to partial melting of the TB crust subducted beneath the Paleo-Tethys accretionary complex (Mitchell, 1977; Nie et al., 2012). If that is the case, the CM granitoids can be interpreted as orogenic and due to the collision of TB with the Yunxian–Jinggu Arc. The earliest CM granitoids (237 Ma) imply that the collisional orogeny (crustal thickening) started during or prior to early Late Triassic. This interpretation is consistent with the suggested closure of the Paleo-Tethys during Late–Middle Triassic.

3.3. The Yunxian–Jinggu granitoid province

The Yunxian–Jinggu granitoid province comprises a series of plutons in the Yunxian–Jinggu Arc Terrane including the Lincang granodiorite, Lincang biotite granites/monzogranites, Banpo and Nanlinshan complexes, Bulangshan muscovite granite, and the Mengsong two-mica granite (Fig. 9). Available radiometric age data for the granites suggest that there was more than one phase of granite intrusion. Two major periods of tectono-magmatism have been recognized in the eastern

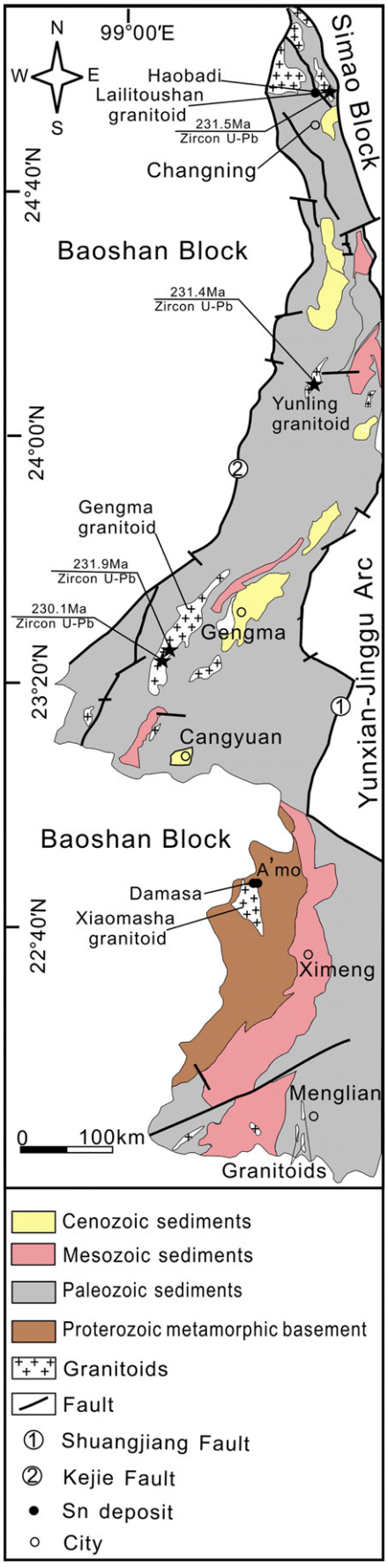
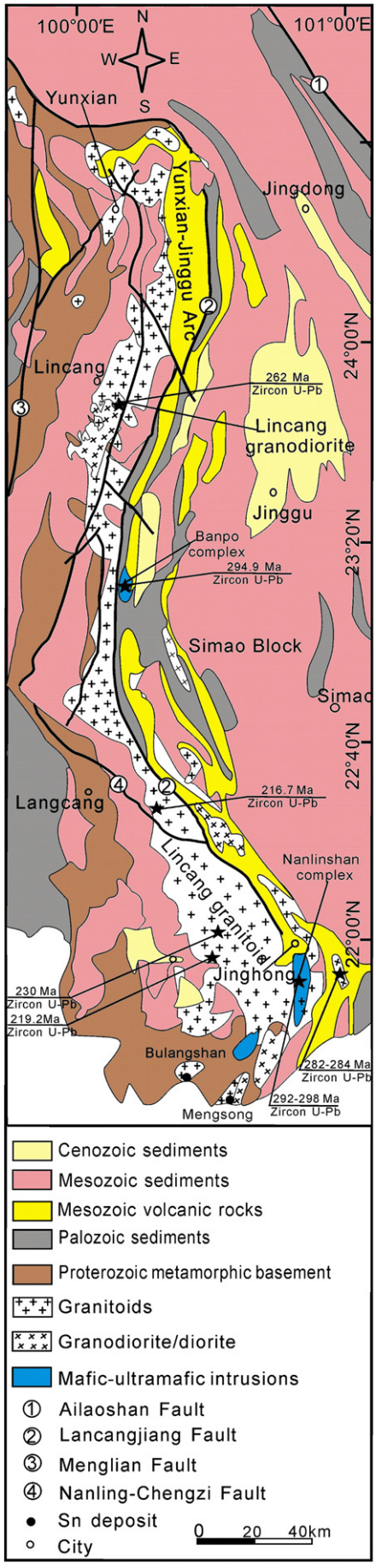


Fig. 8. Simplified geological map showing the granitoids and the distribution of significant Sn deposits in the Changning–Menglian Sn metallogenic belt.
Modified after Nie et al., 2012.



YJ granitoid province, i.e., Permian and Late Triassic. The Permian tectono-magmatism lasted from 298 to 262 Ma, and is represented by the Lincang granodiorite in the middle part of the Lincang granitoids — the Banpo and Nanlinshan complexes (Yu et al., 2003; Hennig et al., 2009; Li et al, 2011, 2012). The Late Triassic to Early Jurassic Lincang biotite granites/monzogranites have ages of 239–189 Ma (Shi et al., 1989; BGMRY, 1990; Qin, 1991).

Geochemical data of the Permian Lincang granitoids from Yu et al. (2003) and Li et al. (2011, 2012) show a compositional range from granodiorite to tonalite (Fig. 4). The rocks have SiO_2 contents ranging from 53.53 to 75.83%, relatively high Al_2O_3 contents (average 14.40%) and K_2O contents (average 3.04%), which define the granitoids as belonging to the calc-alkaline or tholeiitic series (Fig. 5). The average alumina-saturation index (A/CNK molar) of the Permian Lincang granitoids is 1.05. According to the Al vs. (Na + K + 0.5Ca) diagram (Fig. 6b), the Lincang granitoids have characteristics of either I-type granite or S-type granite. The Permian Lincang granitoids have similar chondrite-normalized REE patterns, showing low REE concentrations (average 142.64 ppm), with average chondrite-normalized La/Yb ratio of 10.50 and average chondrite-normalized La/Sm ratio of 5.07. Their chondrite-normalized REE patterns show LREE enrichment (LREE/HREE = 6.70) with negative Eu anomalies ($Eu/Eu^* = 0.72$). They show a narrow range of εNd(t) values (−3.6 to 5.3), whereas their initial $^{87}Sr/^{86}Sr$ (I_{sr}) ratios vary from 0.70410 to 0.70953 (Hou et al., 2008; Yang et al., 2009). These isotopic compositions are consistent with the characteristics of the Permian Lincang granitoids as I-type granites. Based on trace and major element compositions for tectonic discrimination, the Permian Lincang granitoids fall mostly in the volcanic arc field associated with a subduction-related setting (Fig. 7).

Geochemical data of the Late Triassic to Early Jurassic Lincang granitoids (Kong et al., 2012) show that they are dominated by granodiorite and tonalite (Fig. 4). These granitoids have similar major element contents, with average SiO_2 contents of 69.19%, average Al_2O_3 contents of 14.39% and average K_2O contents of 4.17%. These average compositions define the Late Triassic to Early Jurassic Lincang granitoids as members of the high-K (calc-alkaline) series (Fig. 5). The granitoids have an average alumina-saturation index (A/CNK molar) of 1.05 and average A/NK molar of 2.10. These characteristics and their corundum normative character (>1 wt.%) define the Lincang granitoids as peraluminous (Fig. 6a). According to the Al vs. (Na + K + 0.5Ca) diagram (Fig. 6b), the Lincang granitoids have characteristics of S-type granites. These granitoids have similar chondrite-normalized REE patterns, showing high REE concentrations (average 278.52 ppm), average chondrite-normalized La/Yb ratio of 16.16 and average chondrite-normalized La/Sm ratio of 6.16. Their chondrite-normalized REE patterns show LREE enrichment (LREE/HREE = 9.43) with negative Eu anomalies ($Eu/Eu^* = 0.46$), which probably indicate the fractional crystallization of plagioclase. Zircons in the Late Triassic to Early Jurassic Lincang granitoids have negative εHf(t) average value of −12.9, reflecting crustal derivation (Kong et al., 2012). They show old Nd model ages of 2.2–1.9 Ga, indicating derivation from an ancient crustal source (Kong et al., 2012). Based on trace and major element compositions for tectonic discrimination, the Late Triassic to Early Jurassic Lincang granitoids fall mostly in the syn-collision field (Fig. 7).

Because the Permian Lincang granitoids are I-type, they have been regarded to be of magmatic arc origin (Mitchell, 1977; Cobbing et al., 1986; Kamata et al., 2002). The magmatism is considered to have been induced by the eastward subduction of the Paleo-Tethys beneath the margin of the Simao Terrane (Mitchell, 1977). However, the petrogenesis of the Late Triassic–Early Jurassic S-type Lincang granitoids has been explained as due to partial melting of the TB crust subducted

Fig. 9. Simplified geological map showing the granitoids and the distribution of significant Sn deposits in the Yunxian–Jinggu Arc Terrane.
Modified after Kong et al., 2012.

beneath the Paleo-Tethys accretionary complex, which can be interpreted as orogenic and due to the collision of TB with the Yunxian–Jinggu Arc.

4. Spatial distribution of Sn-polymetallic deposits

More than 5% resources of Sn-polymetallic deposits in China formed in the SWST, largest only after the South China Sn-polymetallic metallogenic belt (Zhao and Wu, 2004). The Sn-polymetallic deposits are mainly concentrated in the TB and CM belts, corresponding to the western, and central parts of granitoid provinces in the SWST (Figs. 1, 3, 8). The eastern granitoid province in the SWST does not contain significant Sn deposits but only several mineralized zones, such as the Bulangshan and Mengsong (Fig. 9).

Various attempts have been made to classify primary Sn deposits, such as tectonic setting of the mineral assemblage (Varlamoff, 1975), the nature of intrusion (Materikov, 1977), and igneous activity (Taylor, 1979). Zhao and Wu (2004) proposed the following classifications: (1) deposits associated with S-type granites (including altered granite type, pegmatite type, greisen type, skarn type and hydrothermal vein type); (2) deposits associated with volcanic and subvolcanic rocks; (3) deposits associated with A-type granites; (4) deposits associated with sea floor exhalative mineralization; and (5) placer type deposits. Taylor and Pollard (1986) divided Sn deposits into eight subtypes, namely greisen, skarn, carbonate replacement, vein system, stockwork/brittle fracture, brecciated, pegmatite, and argillic (soft-work). The SWST can be considered to be related to plutonic igneous activities, but typically in an orogenic setting. Thus, the Sn deposits in this belt can be classified using the schemes of Taylor and Pollard (1986) and Zhao and Wu (2004) with minor modifications, such as the altered granite type, pegmatite type, greisen type, skarn type and hydrothermal vein type.

The geological and geochemical characteristics of representative Sn deposits in the SWST are given in Tables 1 and 2.

4.1. TB Sn-polymetallic metallogenic belt

The TB Sn-polymetallic metallogenic belt links to the Western Burma–Thailand belt. The Tengchong area is composed of two large-sized Sn deposits (Xiaolonghe and Lailishan), one large-sized Sn-bearing Nb–Ta–Li deposit (Baihuanao) and five medium-sized Sn deposits in addition to nearly a hundred mineralized occurrences (Fig. 3; Hou et al., 2007). Some skarn Sn deposits (e.g., Diantan) are related to the Early Cretaceous granodiorite and granite along the outer contact zone between granitoid and wallrock, while other hydrothermal vein type mineralization occurs in the inner contact zone. The Baoshan area is composed of Tiechang and Shiganghe the hydrothermal vein type Sn deposits in addition to nearly a ten mineralized occurrences (Fig. 3).

The Xiaolonghe Sn deposit is one of the typical members related to the Late Cretaceous biotite granite in the TB Sn metallogenic belt, western Yunnan (Fig. 3; Liu et al., 1993). The deposit is composed of the hydrothermal vein ores, greisen type ores and subordinate skarn ores. It contains 26,200 t of Sn from ores with 0.18% to 0.42% Sn (Liu et al., 1993). The orebodies vary greatly in size, ranging in length from 30 to 300 m and in thickness from 1 to 20 m. Orebodies are divided into the upper belt and the lower belt. Tin orebodies in the upper belt are stratiform in the contact zone between granite and sandy slate, and the ores are of skarn and greisen types associated with massive mineralization. The tin orebodies in the lower belt are of greisen type, essentially controlled by granite fractures, and typically of vein-type mineralization. The minerals are pyrite, cassiterite, magnetite, topaz, and muscovite. The ores commonly occur as disseminations, and some as irregular veins and veinlets. Wallrock alteration, related to mineralization, includes greisenization, sericitization, silicification, and chloritization. A well-defined zoning pattern is discernible in the orebodies from an interior to an exterior zone with mineral associations such as cassiterite → cassiterite, fluorite, quartz, greisen → cassiterite, biotite, quartz, greisen → cassiterite, topaz, quartz, greisen → topaz, biotite granite → sericitization granite. Magmatic water is dominant in the fluid in the Xiaolonghe Sn deposit. According to Shi et al. (1989), the mineralization process can be divided into two periods — skarn and hypothermal. The skarn phase has a simple mineral assemblage of garnet, diopside, epidote, and magnetite, under two stages: (1) garnet–diopside–magnetite; and (2) epidote–magnetite. The hypothermal period can be divided into three stages: (1) greisen–beryl; (2) greisen–topaz; and (3) K-feldspar–quartz. In the greisen–beryl stage, homogenization temperatures of fluid inclusions in beryl from the greisen type ores range from 455 °C to 469 °C. In the greisen–topaz stage, homogenization temperatures of fluid inclusions in topaz from the greisen type ores range from 345 °C to 435 °C (Shi et al., 1989; Liu et al., 1993).

The Lailishan greisen type Sn deposit is a typical member of the TB Sn metallogenic belt. It contains 42,600 t of Sn from ores with 0.63% to 1.58% Sn (Fig. 3; Liu et al., 1993; Hou et al., 2008). The stratigraphic sequence in the Lailishan deposit is the Carboniferous Menghong Group, which is composed of sandstone and slate. The most prominent structural features are NE-trending faults with moderate dips of 40°–50° toward NW. The major magmatic rocks are the Lailishan biotite granite, monzogranite, and feldspar granite. The Sn orebodies distribute in four areas: Taojinchu, Sanhedong, Laoxiongwo, and Siguangping in the Lailishan deposit. The Taojinchu Sn orebody occurs at the contact zone between feldspar granite and sandstone and slate, which range in length from 33 to 160 m and in thickness from 1.0 to 12.5 m. The Sanhedong Sn orebody occurs in the surrounding fractured zones of the monzogranite, and feldspar granite with 365 m length and 1.3 to 25.7 m thickness. The Laoxiongwo and Siguangping Sn orebodies are controlled by the NE-trending faults in the Carboniferous Menghong Group. The host greisen consists of quartz, muscovite, topaz and fluorite, and accounts for 30–50% of the ore. Sulfides are mainly composed of pyrite, pyrrhotite and lesser amounts of sphalerite and galena, occupying 30–60% of the ore. The ores commonly show disseminated, veined, bedded, massive, and brecciated structures. Wallrock alteration includes potassic alteration, silicification, pyritization, sericitization, kaolinization, and chloritization. The $\delta^{34}S_{CDT}$ values of pyrites in the deposit vary from +3.4 to +6.5‰. They are similar with those of the granite (+2.8‰ to +5.7‰) (Shi et al., 1989), which are pre-ore pyrites formed during normal granite crystallization. For fluids in cassiterite–sulfide and quartz from the Lailishan Sn deposit, estimated δD values range from −125‰ to −83‰ and estimated $\delta^{18}O$ values range between 11.2‰ and 13.7‰ (Hou et al., 2008). According to Shi et al. (1989), the mineralizing process can be divided into skarn and hydrothermal phases. The hydrothermal stage can be divided into four phases: (1) greisen; (2) greisen–pyrite; (3) greisen–quartz; and (4) cassiterite–dnieprovskite. The skarn stage has mineral assemblage of garnet, diopside, magnetite, and quartz. In the greisen stage, homogenization temperature of fluid inclusions in quartz is 490 °C average value. In the greisen–pyrite stage, homogenization temperatures of fluid inclusions in quartz range from 230 °C to 340 °C. In the greisen–quartz stage, homogenization temperatures of fluid inclusions in quartz range from 220 °C to 330 °C. The fourth stage has a simple mineral assemblage of dnieprovskite and cassiterite, with homogenization temperatures of fluid inclusions in quartz from 170 °C to 290 °C (Shi et al., 1989; Liu et al., 1993). Salinities of salt-bearing fluid inclusions are commonly 29.9–34.8 wt.% NaCl eq., whereas salinities of daughter mineral-absent fluid inclusions typically range between 3.4 and 25.0 wt.% NaCl eq. (Shi et al., 1989; Liu et al., 1993; Hou et al., 2008; Feng, 2012).

4.2. Middle Sn metallogenic belt

The CM Sn metallogenic belt is located in the southern segment of the Qiangtang Block (Figs. 1, 8). In this belt and the surrounding region, several fault zones are distributed, such as the Shuangjiang fault zone to the east, and the Kejie fault zone to the west. The CM Sn metallogenic belt comprises two medium-sized hydrothermal vein type deposits

Table 1
Grade, tonnage, granitoid and mineralization ages of significant Sn deposits in the SWST.

Granitoid province	Deposit	Long./Lat.	Deposit type	Tonnage (t)	Grade (%)	Host rocks	Granitoid age (Ma)	Mineralization age (Ma)	Reference
Tengchong–Baoshan	Diantan	98.43/25.65	Skarn type Sn deposit	No data	No data	Diantan granodiorite, dolomitic limestone, mudstone and sandstone	127, biotite Rb–Sr age	No data	Chen (1987), Shi et al. (1991), Liu et al. (1993)
	Dadongchang	98.73/25.53	Hydrothermal vein type Sn deposit	Sn: 10,000	Sn: 0.016–0.26 Pb: 0.67–15.7 Zn: 1–35 Cu: 0.38–1.08	Donghe biotite granite, dolomitic, limestone, arenaceous mudstone and sandstone	Early Cretaceous	118.0, Re–Os molybdenite age	Shi et al. (1989), Dong et al. (2005)
	Baihuanao	98.22/24.83	Greisen type Sn-bearing Nb–Ta–Li deposit	Sn: 12,638 Nb_2O_5: 7078 Ta_2O_5: 2632 Li_2O: 33,824	Sn: 0.014 Nb_2O_5: 0.007 Ta_2O_5: 0.030 Li_2O: 0.033	Xinqi feldspar granite	65.6, biotite Ar–Ar age;	No data	Li et al. (2009), Hou et al. (2008)
	Xiaolonghe	98.43/25.50	Greisen type	Sn: 26,200	Sn: 0.18–0.42	Guyong biotite granite, sandy slate	76.0–67.8, Zircon U–Pb	No data	Shi et al. (1989), Liu et al. (1993)
	Dasongpo	98.34/25.65	Greisen type	Sn: >1000	Sn: 0.13–2.62	Guyong biotite granite, sandy slate	67.8–65.9, K-feldspar Ar–Ar age; 76.0–67.8, Zircon U–Pb	75.5, cassiterite U–Pb	Xia (2003), Feng (2012), Jiang et al. (2012), Ma et al. (2013)
	Lailishan	98.22/24.83	Greisen type	Sn: 42,600	Sn: 0.63–1.58	Lailishan biotite granite, monzogranite, and feldspar granite; sandstone and slate	53.2, Zircon U–Pb; 52.3–50.1, biotite Ar–Ar age	No data	Shi et al. (1989), Liu et al. (1993), Ding (2008), Hou et al. (2008)
	Tiechang	99.15/25.58	Hydrothermal vein type Sn deposit	No data	Sn: 1.22	Tiechang granite, gneiss	No data	No data	Zhang (1996a)
	Shiganghe	89.97/25.74	Hydrothermal vein type Sn deposit	Sn: 2885 WO_3: 6320	Sn: 0.16–0.43 WO_3: 0.22–1.23	Zhibenshan granite and muscovite granite, gabbro-diabase dike, schist, slate, metamorphic sandstones	126.7, Zircon U–Pb	38.6–68.6, muscovite Ar–Ar age; 145.0, cassiterite Ar–Ar age	Zhang (1996b), Qiu et al. (1994), Tao et al. (2009)
Changning–Menglian	Haobadi	99.58/24.61	Hydrothermal vein type Sn deposit	Sn: 8600	Sn: 0.54–1.37	Lailitoushan monzogranite, sandstone, quartz sandstone, and quartzite	231.5, Zircon U–Pb	No data	Shi et al. (1989), Zhao and Wu (2004), Nie et al. (2012)
	A' mo	99/46/22.84	Hydrothermal vein type Sn deposit	No data	Sn: 0.15–2.00	Xiaomasa biotite granite, two-mica granite, and pegmatite, marble, schist	No data	21.5, Rb–Sr taeniolite age	Shi et al. (1989), Zhu et al. (2012)
	Damasa	99.46/22.84	Pegmatite type Sn deposit	No data	No data	Xiaomasa biotite granite	No data	No data	Shi et al. (1989, 1991)
Lincang	Bulangshan	100.45/21.43	Hydrothermal vein type Sn deposit	No data	No data	Bulangshan muscovite granite	228, muscovite Rb–Sr age	No data	Shi et al. (1989)
	Mengsong	100.54/21.38	Hydrothermal vein type Sn deposit	No data	No data	Mengsong two-mica granite	231.3, whole rock isochron age Rb–Sr age	No data	Shi et al. (1989)

Table 2
Geological characteristics and ore-forming fluids of significant Sn deposits in the SWST. The data for each deposit are from references listed in Table 1.

Deposit no.	Tectonic unit	Orebody morphology	Ore structure	Typical minerals	Wall-rock alteration	Temperature (°C)/stage	Salinity (NaCl wt% eq.)
Diantan	Tengchong Block	Lens, stratabound	Massive, disseminated, with some irregular vein, veinlets, and networks	Cassiterite, magnetite, pyrite, galena, sphalerite, pyrrhotite	Skarnization, silicification, sericitization, chloritization	480–600 (skarn)	No data
Dadongchang	Tengchong Block	Vein, stratabound	Disseminated, banded and massive	Pyrite, cassiterite, galena, sphalerite, magnetite, pyrrhotite	Skarnization, silicification, tremolitization, sericitization, chloritization	No data	No data
Bainaohua	Tengchong Block	Vein	No data	Nb, Ta, Li, Cs, Y, Sc minerals, wolframite, cassiterite, columbite-tantalite, pyrite	Albitization, greisenization, amazonitization	No data	No data
Xiaolonghe	Tengchong Block	Irregular	Disseminated, and some irregular vein, veinlets vein	Cassiterite, topaz, columbite–tantalite	Potassic alteration, albitization, greisenization, sericitization	455–469 (greisen–beryl); 345–435 (greisen–topaz)	No data
Dasongpo	Tengchong Block	Irregular	Disseminated, and banded	Cassiterite, pyrite, columbite–tantalite, magnetite	Greisenization, sericitization, silicification	350–420 (greisen–beryl–biotite–quartz); 300–320 (greisen–topaz–muscovite–quartz); 200–240 (pyrite–quartz)	6.8–17.9, with peak at 11–14
Lailishan	Tengchong Block	Vein, stratabound	Disseminated, veined, beded, massive, and brecciated	Cassiterite, topaz, sulfides	Potassic alteration kaolinization, sericitization, silicification, pyritization, chloritization	490 (greisen); 230–340 (greisen–pyrite); 220–330 (greisen–quartz); 170–290 (cassiterite–dnieprovskite)	3.4–34.8
Tiechang	Baoshan Block	Vein	Disseminated, massive, veined, beded, and brecciated	Cassiterite, arsenopyrite, pyrite, pyrrhotite	Sericitization, silicification, pyritization, chloritization	No data	No data
Shiganghe	Baoshan Block	Stratiform, vein	Massive, disseminated, with veined and network	Cassiterite, beryl, wolframite, arsenopyrite, pyrite, pyrrhotite	Silicification, albitization, greisenization, tourmalinization	405–450 (cassiterite–wolframite–tourmaline–quartz); 270–380 (cassiterite–wolframite–sulfide)	No data
Haobadi	Baoshan Block	Stratiform, lenticular	Disseminated, brecciated, and network	Pyrite, cassiterite, arsenopyrite, pyrrhotite	Silicification	415–460 (tourmaline); 345–396 (cassiterite–tourmaline–quartz); 275–321 (cassiterite–sulfide)	No data
A' mo	Changning–Menglian Suture	Stratiform, lenticular, vein	Disseminated, massive, veined, and brecciated	Pyrite, cassiterite, scheelite, bismuthinite columbite–tantalite	Albitization	450–470 (pegmatite); 430–450 (cassiterite–tourmaline–quartz); 350–380 (cassiterite–quartz); 330–350 (cassiterite–arsenopyrite–taeniolite); 280–350 (cassiterite–sulfide)	3.0–38.0
Damasa	Changning–Menglian Suture	Vein, lenticular,	Massive, beded, and disseminated	Cassiterite, topaz, columbite–tantalite, beryl	No data	480–600 (pegmatite)	No data
Bulangshan	Yunxian–Jinggu Arc	Vein	Massive, disseminated, with veined and network	Pyrite, cassiterite, arsenopyrite, pyrrhotite	Albitization, tourmalinization, silicification, greisenization	No data	No data
Mengsong	Yunxian–Jinggu Arc	Vein	Massive, disseminated, with veined and network	Pyrite, cassiterite, arsenopyrite, pyrrhotite	Albitization, tourmalinization, silicification, greisenization	No data	No data

(Haobadi and A' mo) and one medium-sized pegmatite type deposit (Damasa).

The Haobadi Sn deposit is located in the northern segment of the CM Sn metallogenic belt (Fig. 8), and the Lailitoushan monzogranite pluton occurs at ca. 0.5 km NW of this deposit. The stratigraphic sequence mainly includes Cambrian schist, Devonian phyllite and slate, Triassic clastic rocks, Cenozoic sandy conglomerate and alluvium. The Haobadi Sn deposit is hosted in Triassic clastic rocks composed of sandstone, quartz sandstone, and quartzite, with minor carbonaceous shale and limestone. The deposit is localized along the NW–NWW-trending faults and contains 8600 t of Sn from ores with 0.54% to 1.37% Sn. The orebodies are stratiform and lenticular in shape, which belong to the hydrothermal vein Sn deposit. The orebodies vary greatly in size, ranging in length from 120 to 800 m and in thickness from 0.40 to 8.13 m (Liu et al., 1993). The ore minerals are cassiterite, pyrite, arsenopyrite, and pyrrhotite, with minor wolframite and galena. The gangue minerals are dominated by quartz and tourmaline. The ores commonly show disseminated, brecciated, and network structures. Wallrock alteration is mainly silicification. According to Liu et al. (1993), the mineralizing process can be divided into three stages: (1) tourmaline, with the homogenization temperatures of fluid inclusions in quartz ranging from 415 °C to 460 °C; (2) cassiterite–tourmaline–quartz, with the homogenization temperatures of fluid inclusions in quartz ranging from 345 °C to 396 °C; and (3) cassiterite–sulfide, with the homogenization temperatures of fluid inclusions in quartz ranging from 275 °C to 321 °C.

The A' mo Sn deposit is located in the southern segment of the CM Sn metallogenic belt (Fig. 8). Here, the Ximeng Group Proterozoic metamorphic rocks consist of biotite plagioclase schist, phlogopite marble and two-mica schist. The Xiaomasa is a composite granitoid comprising biotite granite, two-mica granite, and pegmatites. The A' mo Sn orebodies are controlled by ring and radiating structures (Zhu et al., 2013). The orebodies are stratiform, lenticular, and vein in shape, and belong to the hydrothermal vein Sn deposit. The orebodies range in length from 180 to 250 m and in thickness from 2 to 35 m (Liu et al., 1993). The major ore types in the deposit are cassiterite–tourmaline–quartz and cassiterite–quartz, and the minor ore types are cassiterite–arsenopyrite–taeniolite, cassiterite–sulfide, and cassiterite–pegmatite. The ore minerals are mainly cassiterite, stannite, pyrite, scheelite and bismuthinite, with chalcopyrite, sphalerite and galena. The gangue minerals are dominated by quartz, tourmaline, and muscovite. The ores commonly show disseminated, massive, veined, and brecciated structures. Wallrock alteration is characterized by albitization, tourmalinization, greisenization, and silicification. According to Shi et al. (1989), pegmatite and hydrothermal periods at A' mo Sn deposit were identified. The former consists mainly of cassiterite, thorite, columbite–tantalite, and manganotantalite, with ore-forming temperatures from 450 °C to 470 °C. The latter can be divided into four stages: stage 1 cassiterite–tourmaline–quartz, with ore-forming temperatures from 430 °C to 450 °C; stage 2 cassiterite–quartz, with ore-forming temperatures from 350 °C to 380 °C; stage 3 cassiterite–arsenopyrite–taeniolite, with ore-forming temperatures from 330 °C to 350 °C; and stage 4 cassiterite–sulfide, with ore-forming temperatures from 280 °C to 350 °C. Salinities of salt-bearing fluid inclusions are commonly 29.5–38.0 wt.% NaCl eq., whereas salinities of the daughter mineral-absent fluid inclusions typically range between 3.0 and 10.0 wt.% NaCl eq. (Shi et al., 1989; Liu et al., 1993; Hou et al., 2008; Feng, 2012). The $\delta^{34}S_{CDT}$ values of sulfide minerals range from +3.5 to +7.2‰ (Shi et al., 1989). These rather uniform values point to a large-scale hydrothermal system with a homogeneous source of sulfur, which was likely of magmatic origin (Shi et al., 1989). It should be noted that the A' mo sulfides have slightly higher $\delta^{34}S_{CDT}$ values than some other tin deposits, implying an initial relative enrichment in the $\delta^{34}S_{CDT}$ content of the sulfur source. This is not unexpected for S-type granitic magmas, which acquire most of their sulfur through assimilation of country rocks.

5. Discussion

5.1. Deposit types

Examples for the hydrothermal vein type deposits include the Haobadi, A' mo, Tiechang, Shiganghe, and Dadongchang ore deposits. Next most abundant are greisen type, skarn type, and pegmatite type deposits, and the examples include the Lailishan and Xiaolonghe greisen deposits, Diantan skarn deposit, and the Damasa pegmatite-hosted deposits. Other types of Sn deposits are rare in the SWST.

5.2. Mineralogy

Most pegmatite type Sn deposits in the SWST have a simple mineral assemblage consisting primarily of cassiterite, topaz, beryl, columbite–tantalite, xenotime, aplite, apatite, zircon, tourmaline, and almandine.

Greisen type Sn deposits in the SWST occur in various settings with respect to granite intrusive, for example deposits near the roof zones of granites (e.g., Lailishan deposit) or deposits protruding from the granite into the country rock (e.g., Xiaolonghe deposit). The mineral assemblages of greisen type Sn deposits near the roof zones of granites can be divided into the following seven stages (Shi et al., 1989; Shi et al., 1991), from earliest to latest: (1) cassiterite–lithium–bearing biotite–topaz–quartz; (2) cassiterite–muscovite–topaz–quartz; (3) cassiterite–topaz–quartz; (4) cassiterite–beryl–muscovite–quartz; (5) cassiterite–quartz; (6) cassiterite–potash feldspar; and (7) cassiterite. Topaz minerals are abundant in the early stages, whereas only minor amounts are present in the later stages. The mineral assemblages in the country rocks may appear similar, to varying degrees, to those near the roof zones of granites to varying degrees, but they also show distinct features such as pyrite and other sulfides. The hydrothermal vein type Sn deposits can have variable mineral assemblages, such cassiterite–tourmaline–quartz, cassiterite–quartz, cassiterite–arsenopyrite–pyrite, or cassiterite–quartz–pyrrhotite–pyrite. Most of the hydrothermal vein type Sn deposits contain only subordinate amounts of Cu, Pb, Zn sulfide minerals or none at all, the dominant iron minerals being pyrite and pyrrhotite.

In skarn type Sn deposits, garnet, wollastonite, pyroxene, vesuvianite and/or forsterite are the major silicates formed in the early stages of mineralization. Four sub-types of skarn deposits have been identified based on the major early-stage minerals: magnetite–tin, andradite–wollastonite, magnetite–fluorite–vesuvianite, and forsterite–pyroxene–spinel. Sulfide minerals occur in the late stages in skarn Sn deposit, such as pyrite, galena, and sphalerite.

5.3. Nature of ore fluids

The fluid inclusion data from tin deposits in the SWST are given in Table 2, and plotted in Figs. 10 and 11. Only one of the listed deposits is pegmatite-hosted (Damasa), one is skarn mineralization (Diantan), and three deposits are dominated by greisen type mineralization (Xiaolonghe, Lailishan, and Dasongpo) in addition to hydrothermal vein type mineralization. Salinities of salt-bearing fluid inclusions are commonly more than 29 wt.% NaCl eq., whereas salinities of daughter mineral-absent fluid inclusions typically range between 3 and 25 wt.% NaCl eq. Early-stage fluid inclusions, which are more likely representative of fluids that deposited cassiterite, tend to show higher homogenization temperatures than the late-stage fluids. Homogenization temperatures of early-stage fluid inclusions are mainly in the 480–600 °C range for pegmatite and skarn type Sn deposits, 300–490 °C range for greisen type, and 270–450 °C range for the hydrothermal vein type Sn deposits (Fig. 11; Shi et al., 1989, Shi et al., 1991; Liu et al., 1993; Schwartz et al., 1995; Xia, 2003; Ding, 2008; Hou et al., 2008; Feng, 2012). Homogenization temperatures of early-stage fluid inclusions are mainly in the 275–350 °C range for greisen type, and 170–340 °C range for

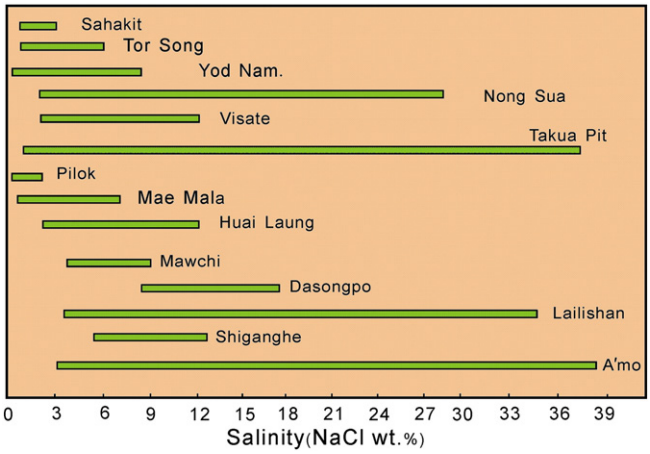


Fig. 10. Summary of salinity data from selected Sn deposits based on fluid inclusion studies in the SWST and its adjacent areas (Shi et al., 1989; Liu et al., 1993; Schwartz et al., 1995; Hou et al., 2007).

the hydrothermal vein type Sn deposits (Shi et al., 1989, Shi et al., 1991; Liu et al., 1993; Hou et al., 2007).

5.4. Geodynamic constraints on Sn mineralization

During Late Devonian–Early Permian, the northern margin of the Gondwana supercontinent was dismembered and a series of small continental fragments developed such as the Qiangtang Block and the Simao Block (Sengor, 1987; Metcalfe, 1996, 1997; Ueno et al., 2003; Wang et al., 2008, 2010a, 2010b; Deng et al., 2011a, 2011b). This event resulted in the opening of the Meso-Tethys (e.g., Bangong–Nujiang ocean) and subduction of the Paleo-Tethys (Fig. 12a) (Metcalfe, 1996, 2002; Bortolotti and Principi, 2005). Block separation and subduction of the Paleo-Tethys ocean were accompanied by volcano-plutonic activities such as the tholeiites in the Baoshan Terrane, the western CM ocean basin, and the Jinghong back-arc basin (Ueno and Hisada, 2001; Sone and Metcalfe, 2008). During Early Permian, the CM Ocean was subducted beneath the Simao Block, resulting in the formation of Yunxian–Jinggu arc along the block, especially the Lincang I-type granodiorite in the middle part of the Lincang granitoids (Fig. 12a). The Late Permian was an important stage of closure and collapse of the Jinghong back-arc oceanic basin (Fig. 12b). In Late Triassic, the Paleo-Tethys was completely closed in the SWST. During the Middle Triassic–Early Jurassic, the collision and amalgamation of the TB and Simao Blocks resulted in the formation of the CM Suture Zone and the emplacement of S-type granitoids (Fig. 12c; Metcalfe, 2000; Sone and Metcalfe, 2008) such as the Lincang biotite granites/monzogranites, Bulangshan muscovite granite, Mengsong two-mica granite, Xiaomasa biotite granite, and two-mica granite. These S-type granitoids were formed by partial melting of the Simao crust (Cobbing et al., 1986; Charusiri et al., 1993; Metcalfe, 2000; Sone and Metcalfe, 2008). The CM and YJ Sn deposits are mainly related to these S-type granitoids, such as the Damasa pegmatite type Sn deposit, and the Bulangshan and Mengsong hydrothermal vein type Sn deposits. After the subduction of the Bangong–Nujiang Tethyan Ocean lithosphere, during the Permian–Early Cretaceous, the south-eastward-moving west Burma Terrane collided with the TB Terrane. This collision caused intense shortening and double thickening of the crust in the interior of the TB Terrane, as well as the formation of the Bangong–Nujiang Suture Zone and the widespread emplacement of Early Cretaceous S-type granitoids (Fig. 12d), such as the Diantan granodiorite, Donghe biotite granite, Zhibenshan granite and muscovite granite. Widespread Sn mineralization in the TB Block was associated with the S-type magmatism, such as the Diantan skarn type Sn deposit, and Tiechang and Shiganghe hydrothermal vein type Sn deposits.

After the subduction of the Indus–Yarlung Zangbo Tethyan mid-ocean ridge during 100–80 Ma, the northward-moving Indian continent collided with the Asian continent (Chatterjee et al., 2012). This collision led to the formation of the Himalayan Orogen, which caused intense shortening and double thickening of the crust in the interior of the TB, as well as the formation of the Indus–Yalung Zangbo Suture and the extensive belts of granitoids (Fig. 12e), such as the Lailishan biotite granite, monzogranite, and feldspar granite. Despite earlier controversies about the timing of the initial collision, most researchers now consider that the collision probably occurred in the Early Cenozoic, during 65–55 Ma (Searle et al., 1987; Yin and Harrison, 2000; Dupont-Nivet et al., 2010; Liebke et al., 2010; Yi et al., 2011). Therefore, the most important period for Sn deposit formation, which was during the Late Cretaceous–Early Cenozoic era, broadly corresponds to the India–Asia continental collision. The Xiaolonghe and Lailishan greisen type Sn deposits in the TB belt formed during that period.

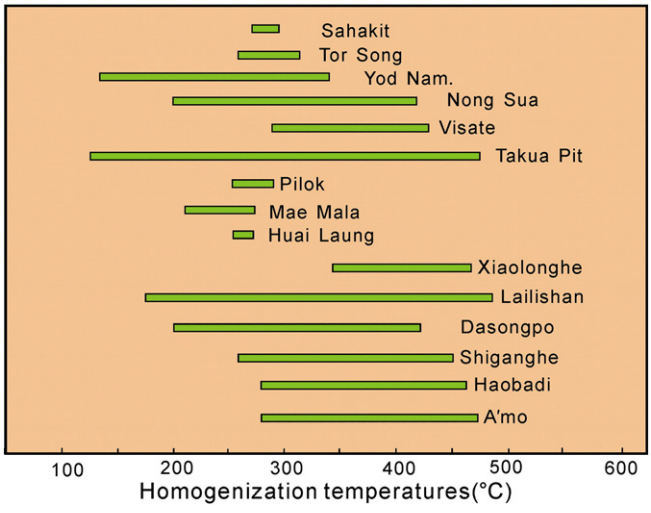


Fig. 11. Summary of homogenization temperatures for selected tin deposits based on fluid inclusion studies in the SWST and its adjacent areas (Shi et al., 1989; Liu et al., 1993; Schwartz et al., 1995; Hou et al., 2007).

6. Summary

1) The Permian (298–262 Ma) I-type granitoids include Lincang granodiorite in the middle part of the Lincang granitoids and the Banpo and Nanlinshan complexes. The Permian magmatism is considered to have been induced by the eastward subduction of the Paleo-Tethys beneath the margin of Simao Terrane. These I-type granitoids apparently do not contain significant Sn deposits.

2) The Middle Triassic–Early Jurassic S-type granitoids are distributed in Changning–Menglian Sn metallogenic belt (with ages of 237–178 Ma) and in the Yunxian–Jinggu Arc Terrane (with ages of 239–189 Ma). The subduction of the Bangong–Nujiang Meso-Tethys and the collision between TB and Simao Blocks resulted in the formation of S-type granitoids. There were widespread hydrothermal vein type and pegmatite type Sn mineralizations related to these S-type granitoids.

3) The Early Cretaceous (136–113 Ma) S-type granitoids in the Tengchong–Baoshan Block resulted from its collision with the western Burma Block, which caused double thickening of the crust and the formation of the hydrothermal and skarn type vein type Sn deposits.

4) The Late Cretaceous–Early Cenozoic (89–52 Ma) S-type granitoids in Tengchong–Baoshan Block likely resulted from the India–Asia continental collision, and the Xiaolonghe and Lailishan greisen type deposits formed during that period.

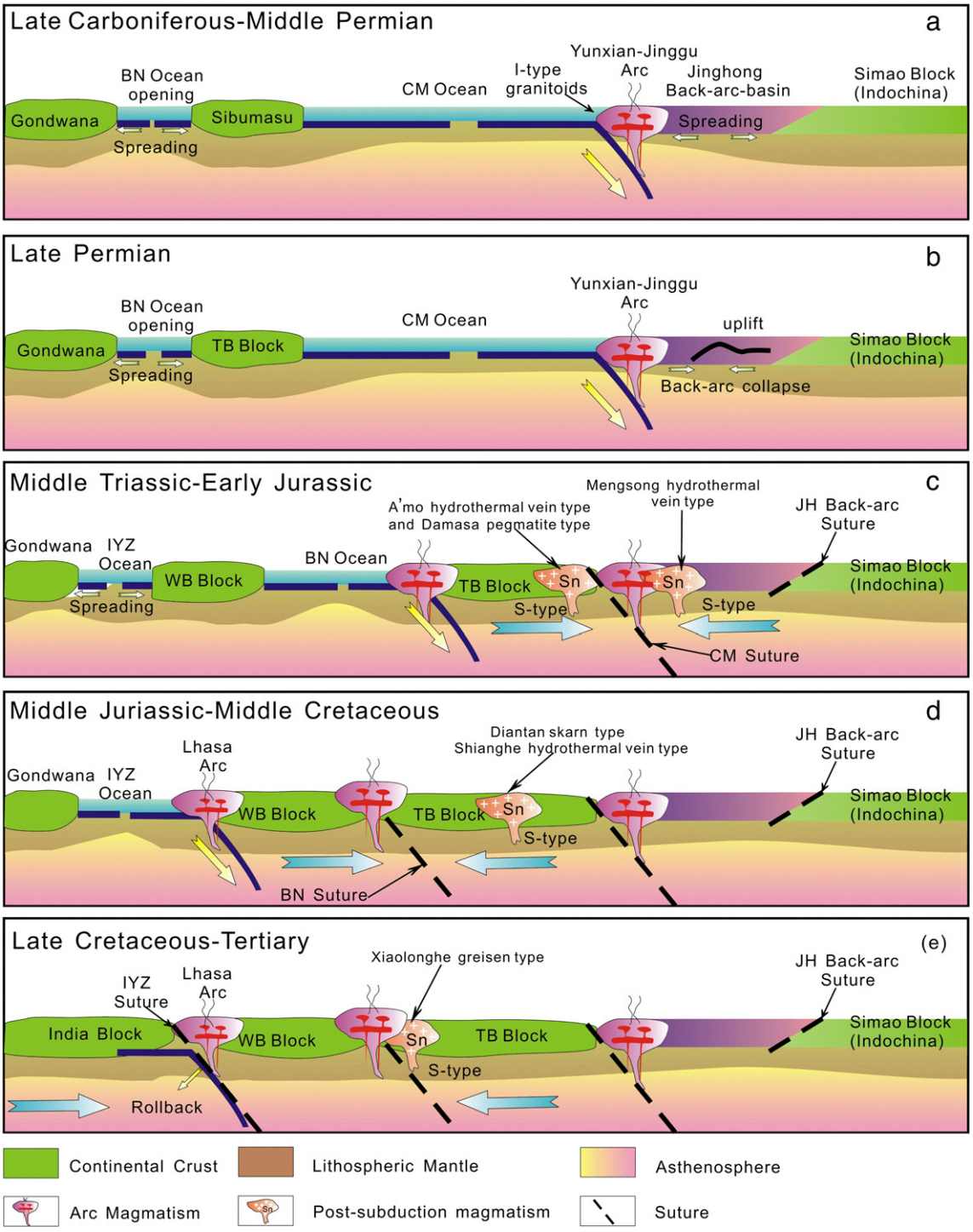


Fig. 12. Tectonic evolution and Sn metallogenesis associated with granitoids in the SWST. (a) During the Late Carboniferous–Middle Permian, the subduction of the CM Paleo-Tethys resulted in the Yunxian–Jinggu volcano-plutonic arc and the Jinghong back-arc basin; (b) during the Late Permian, the Jinghong back-arc collapse; (c) during the Middle Triassic–Early Jurassic, the subduction of the BN Meso-Tethys, and the collision between TB and Simao Blocks resulted in the formation of S-type granitoids related to the pegmatite and hydrothermal vein type Sn deposits; (d) during the Middle Jurassic–Middle Cretaceous, the collision between WB and TB Blocks resulted in the formation of S-type granitoids related to the skarn and hydrothermal vein type Sn deposits; (e) during the Late Cretaceous–Tertiary, the India–Asia continental collision resulted in the formation of S-type granitoids associated with the greisen type Sn deposits.

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